Rayyan Moosani

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Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science — *Expected 2029*

Experience

Software Engineer — May 2026 - Present
XORBIX

- Cut validation time **50x** to **442K rows/sec** via chunked vectorized **pandas** boolean-mask engine
- Slashed **Flask REST API** latency **5.5x** (3.1s to 0.56s) by collapsing 301 **N+1** queries into 3 bulk reads
- Reduced AI extraction time by **95%** (6 min to 5s) in a **C/.NET WPF** app via real-time data streaming
- Generated **200+** files of **.NET/React** apps, cutting setup effort **85%**, via self-made 6-step codegen application

Software Automation Developer — Aug 2024 - Sep 2024
SCITRA

- Built real-time reporting system using **Python Telegram** and **Google Sheets APIs**, reducing latency by **40%** across **30+ lines**
- Engineered **Unity** and **C#** digital twins with **IoT** data, improving predictive maintenance accuracy by **30%**
- Automated fault logging using **Python** pipelines, accelerating engineering resolution time by **25%**

Software Developer — Sep 2025 – Dec 2025
IICUW

- Optimized frontend performance using ES6+ refactoring, reducing bundle size by **20%** and improving load time
- Developed **Python Discord** bots, eliminating manual workflows and increasing efficiency by **35%**
- Built real-time **React** features using event-driven architecture, increasing engagement for **100+ users**

IT Intern — Jun 2024 - Jul 2024
AGMC

- Digitized workflows using **MS Forms** and **SharePoint**, reducing manual processing effort by **30%**
- Migrated legacy systems to **M365**, improving data accessibility and retrieval speed by **40%**
- Resolved **50+ IT tickets** using enterprise tools, improving system reliability and user productivity

Projects

StoryVerse | *TypeScript, Node.js, Express, React, PostgreSQL, Drizzle ORM, FFmpeg, OpenAPI, Zod*

- Shipped a **portal** where viewers watch AI trailers made by authors and **vote to greenlight** them for **funding**.
- Built **AI pipeline** turning scripts into trailers and shorts across **7+ tiers** of **20+ models**.
- Cut generation cost **85%** by building an **editable**, **human-in-the-loop** pipeline state machine.
- Raised **character consistency ∼2/9 to 9/9** via **prompt-injection** layer locking descriptions.
- Synced multi-scene **narration** and **lip-sync** to video via **FFmpeg** with **R2**-backed storage.

Circuit Guard | *Python, FastAPI, Next.js, React Three Fiber, Claude* — **Winner**, FormulaTechHacks 2026

- Built an **AI Model** that spots F1 crashes and tells officials which flag to show, quoting the F1 rulebook
- Mapped **600+** real race incidents to show which corners endanger fans, with a low-cost fix for each
- Pinpointed the 3 corners behind **47%** of serious crashes so track owners can negotiate cheaper **insurance**

Games.Random | *Node.js, Express, MongoDB, Claude API, Docker*

- Built a gaming/learning platform using **Node.js** to generate playable 2D games from text with editable code.
- Streamed live code generation using **Server-Sent Events** to achieve real-time rendering.
- Engineered secure auth & deployment using **MongoDB**, **OAuth 2.0**, and **Docker** containers.

Technical Skills

Languages: Python, C#, TypeScript/JavaScript, SQL, C, Racket, XAML, HTML5, CSS3.
AI/ML: LLMs, RAG pipelines, Anthropic Claude API, OpenAI API, Ollama, LM Studio, Claude Code, prompt engineering, Whisper, Pandas, NumPy, PHOIBLE, Selenium.
Frameworks, Infrastructure & APIs: .NET 8, ASP.NET Core, WPF (MVVM), EF Core, FastAPI, Flask, React, Vite, Node.js, Express, TailwindCSS, Docker, PostgreSQL (pgvector), SQLite, Supabase, MongoDB, REST APIs, WebSockets, OAuth 2.0, xUnit, Azure DevOps, CI/CD (Jenkins), Git/GitHub.